Final Documentation & Project Summary

Project: Stretto Technologies Enterprise Database System

Student Name: GBAGI OMAFUVWE ERIC

Student ID: 301918771

GitHub Repository Link: https://github.com/EricGbagi/advanced-database-portfolio-MIT8103-Omafuvwe-Gbagi

1. Executive Summary & Selected Organisation

1.1 Organisational Context and Business Domain

Stretto Technologies is a Lagos-based e-commerce and electronics sourcing platform specializing in the procurement and retail of high-value consumer gadgets, including smartphones, laptops, and gaming consoles. Rather than maintaining traditional, static warehouse stock that incurs heavy holding costs, the company operates on a dynamic, on-demand sourcing model. When a customer initiates a purchase, inventory is procured instantly from established, trusted vendor channels such as *OA Plug Device*, *UK Direct Importers*, and *Gaming Hub NG*.

1.2 Enterprise Architecture Objectives

To support this commercial model, this project engineered a comprehensive, multi-paradigm enterprise database solution spanning five core technical modules. The objective was to design a robust data architecture that satisfies strict financial accountability while providing the flexibility required for rapid inventory scaling and distributed cloud availability.

The multi-model implementation addresses specific engineering challenges across the business lifecycle:

- **Relational Core (PostgreSQL):** Designed to handle strict ACID transactions, multi-item customer invoicing, granular line-item sourcing, and financial payment reconciliation across a high-volume dataset of 50,000 transactional records.
- **Query Optimization:** Leveraged advanced analytical SQL queries and B-Tree indexing (EXPLAIN ANALYZE) to reduce retrieval latency on time-sensitive transactional reports and revenue summaries.
- **Concurrency & Transactions:** Implemented row-level locking (SELECT ... FOR UPDATE) and serializable isolation levels to prevent concurrency anomalies, lost updates, and phantom reads during concurrent payment processing.
- **NoSQL Document Model (MongoDB Atlas):** Deployed a schema-less document architecture to seamlessly absorb variable product specifications (e.g., IMEI tracking, storage variations) and provide denormalized read performance.

- **Distributed Cloud Architecture:** Configured a fully managed cloud cluster utilizing native replica sets (Primary/Secondary node distribution for high availability), sharding (customer_id horizontal routing), and CAP Theorem CP consistency enforcement.

2. Summary of Portfolio Evidence

The complete technical artifacts, physical SQL schemas, execution plans, transaction logs, and cloud execution proofs are organized within the project repository across five dedicated deliverables:

1. **Portfolio 1 (portfolio-1-database-design.pdf):**

Establishes the foundational database structure. It includes the comprehensive case study description, a fully normalized 6-table relational schema up to the Third Normal Form (3NF), an Entity-Relationship Diagram (ERD) mapping explicit primary-foreign key relationships, explicit constraint definitions, and high-volume mock data generation scripts utilizing PostgreSQL's generate_series function to populate 50,000 transactional records.

2. **Portfolio 2 (portfolio-2-query-optimisation.pdf):**

Demonstrates advanced data retrieval and performance tuning. It features three complex analytical queries—date-range transaction filtering, supplier gross revenue aggregation using multi-table joins and aggregate functions, and outstanding balance reconciliation via LEFT JOIN and COALESCE. It includes baseline execution plans (EXPLAIN ANALYZE), B-Tree index creation on high-cardinality timestamp attributes, and post-index optimization analysis showing transition paths to Bitmap Heap Scans.

3. **Portfolio 3 (portfolio-3-transactions-concurrency.pdf):**

Proves transactional integrity and concurrency control mechanisms. It documents ACID compliance through successful COMMIT execution and constraint-driven ROLLBACK handling upon NOT NULL violations. It further explores row-level locking to mitigate update anomalies and implements strict SERIALIZABLE isolation levels for accurate financial auditing.

4. **Portfolio 4 (portfolio-4-nosql.pdf):**

Evaluates advanced data models by implementing a Document Store via MongoDB Atlas. It outlines the schema suitability for variable electronics specifications, provides a nested JSON document structure representing a complete Stretto Technologies transaction, and supplies implementation evidence including document insertion proofs and filtered projection queries.

5. **Portfolio 5 (portfolio-5-distributed-cloud.pdf):**

Details distributed cloud database deployment. It reviews cloud migration benefits, high availability through replica set health (Primary and Secondary node management), horizontal partitioning via sharding strategies, CAP Theorem trade-offs favoring CP consistency, and live cloud platform execution evidence.

3. Individual Reflection

Working on the Stretto Technologies database architecture provided profound, hands-on insight into the balancing act required in modern data engineering. Moving from strict relational normalization in PostgreSQL to schema-less document modeling in MongoDB Atlas highlighted that no single database paradigm fits every business requirement.

While PostgreSQL excelled at enforcing financial consistency, maintaining referential integrity, and handling complex multi-table joins for financial audits, MongoDB provided unmatched flexibility for handling variable product catalogs without schema migration overhead. Furthermore, configuring replica sets and sharding in MongoDB Atlas demystified the operational complexities of cloud scalability and high availability. Overall, this project bridged the gap between theoretical database design and practical enterprise implementation, sharpening my skills in query tuning, transaction safety, and cloud infrastructure management.

4. References & AI-Use Declaration

References

1. PostgreSQL Global Development Group. (2026). *PostgreSQL 16 Documentation: Concurrency Control, Performance Tips, and SQL Language Reference*. Retrieved from https://www.postgresql.org/docs/16/index.html
2. MongoDB, Inc. (2026). *MongoDB Manual: Document Database Design, Replica Sets, and Atlas Cloud Services*. Retrieved from https://www.mongodb.com/docs/manual/
3. Elmasri, R., & Navathe, S. B. (2015). *Fundamentals of Database Systems* (7th ed.). Pearson.
4. Coronel, C., & Morris, S. (2018). *Database Systems: Design, Implementation, & Management* (13th ed.). Cengage Learning.

AI-Assistance Declaration

In compliance with academic integrity guidelines, generative AI (Gemini) was utilized collaboratively as a personal assistant throughout this project to brainstorm the Stretto Technologies case study parameters, debug SQL syntax, and format Markdown documentation. All database design logic, code execution, query tuning, and final implementations were verified and executed by the author.